

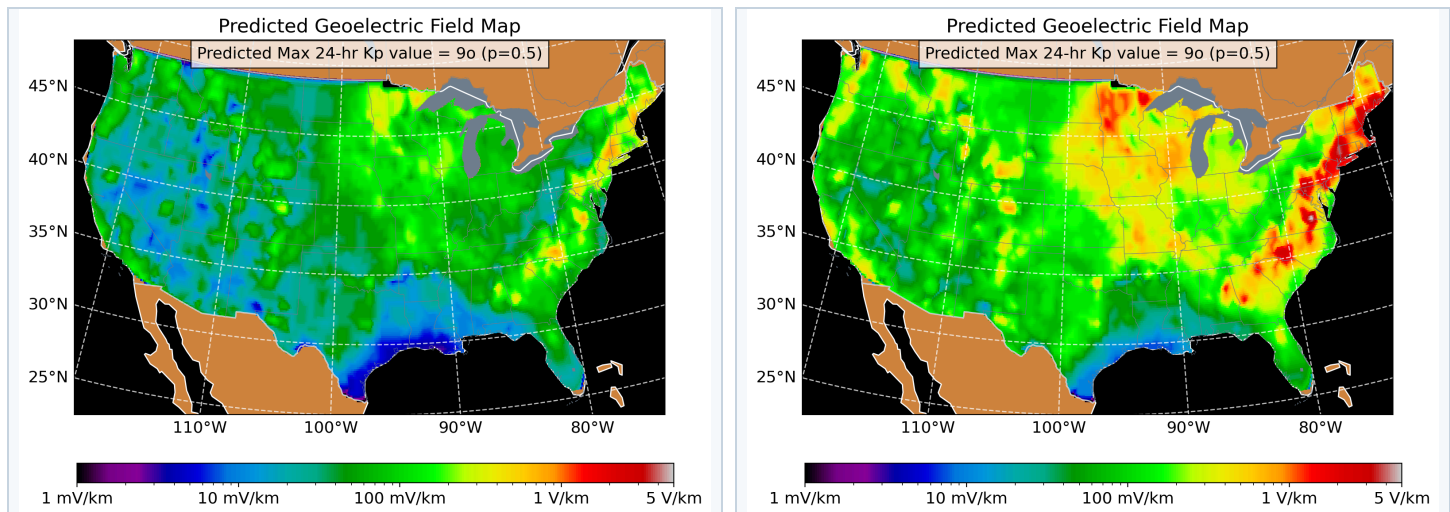


Geomagnetic Disturbance Expected Today into Saturday

This briefing is tailored for reliability coordinators, transmission operators, ISOs/RTOs, and operational coordination groups.

Multiple coronal mass ejections are expected to influence Earth beginning later today. There is an 85% chance of G3 or greater geomagnetic conditions during the Warning period, including a 5% chance of G5 conditions. Power-system response will depend on storm duration, magnetic-field orientation, regional ground conductivity, transmission-system configuration, and operating posture. Operators should prepare for heightened monitoring through the primary arrival window and be prepared for repeated or prolonged periods of elevated geomagnetic activity into the weekend.

PREDICTED REGIONAL GEOELECTRIC-FIELD RESPONSE



Map on the left represents the most-likely values and the map on the right represents the maximum possible values, based on climatological frequencies of geoelectric fields during storm's periods of the predicted k_p value.

HOW TO USE THIS TOOL

- Localized areas of enhanced geoelectric field are possible across portions of the northern and central United States and the Northeast.
- Observed values may vary substantially over short distances due to regional geology and model resolution constraints.
- The map represents a plausible response for the forecast storm level; it is not a prediction of specific transformer currents, voltage problems, or equipment impacts.
- Local GIC measurements and system response should remain the primary operational indicators once the disturbance arrives.

FORECAST CONCERNS & UNCERTAINTY

- Operators should plan for the possibility of elevated geomagnetic activity continuing for 24 to 48 hours, with repeated or renewed disturbance periods possible as additional coronal mass ejections arrive.
- Confidence is high that significant geomagnetic activity will occur, but the magnitude and geographic distribution of grid response cannot be determined before arrival. The largest uncertainty is the strength and duration of southward interplanetary magnetic field after each coronal mass ejection reaches Earth. Multiple arrivals may also prolong or renew the disturbance.

POTENTIAL GRID-RELEVANT IMPACTS

- Voltage alarms or voltage control issues.
- Increased geomagnetically induced current potential.
- Weak to moderate power system fluctuations.
- Greater concern if severe storming is sustained over multiple hours.
- Possible repeated periods of enhanced geomagnetic activity as additional CME structures arrive.

WHAT OPERATORS SHOULD WATCH

- SWPC Warning Updates.
- Observed Space Weather Alerts, especially G4 or G5 conditions.
- Internal GIC monitors, transformer response, voltage alarms, and local system indicators.
- Any reports from neighboring systems or reliability coordination channels.

ACTIVE PRODUCTS

- Space Weather Warning: In effect.
- Observed Space Weather Alerts: Expected if thresholds are reached.

NEXT UPDATE

SWPC will continue issuing Warning Updates at least every 12 hours while the Warning remains in effect, or sooner if observations, impacts, or forecast guidance change.

For additional information, monitor spaceweather.gov and SWPC decision-support guidance.